Six mixed incidence components span a 54-dimensional Hecke floor kernel

Review qualification, 2026-10-30. This is an exploratory relation-stage record. The protocols and results were first published together in the reviewed PR; no verifiable earlier public protocol has been established. Agreement with a recorded forecast is not independently preregistered confirmation. The sealed original report retains its original wording and hashes as history; this reviewed report supplies the current interpretation. Exact finite counts, ranks and proof obligations are unchanged.

The exploratory D_K=-2631, conductor-6 class on X_0(11)/F_23743 has 768 rational points, 576 verified Hecke rows, exact rational rank 480, and nonfloor projected rank 426. Its 54-dimensional floor-only row space has an explicit integer basis: six disjoint alternating degree-2 cycles of 48 rows and 48 degree-3 row-pair differences. Each cycle meets eight pairs, and all 48 overlaps contain both sign ratios. These quantities agree with the recorded forecast for a mixed incidence configuration with six components; prior public preregistration has not been established.

Quantity	Recorded forecast	Checked result
Points at conductors 1,2,3,6	96,96,288,288	96,96,288,288
Degree-2 / degree-3 complete rows	384 / 192	384 / 192, all group-verified
Full rational relation rank	480	480
Nonfloor projected rank	426	426
Signed degree-2 cycles	6	6 of length 48
Degree-3 floor pair differences	48	48 of support size 6
Mixed incidence components / balanced components	6 / 0	6 / 0
Floor-only row-space dimension	54	54

The sealed historical report states that the selection rule was saved before candidate screening. No earlier committed and published protocol is established here; local file ordering is insufficient for preregistered confirmation, so this case is classified as exploratory. Among 30 screened discriminants satisfying its inexpensive arithmetic conditions, the first eligible value was D_K=-2631; the selection receipt retains all rejections and its class-action data. The norm-2 class orders are 16 at conductor 1 and 48 at conductor 3; the ramified norm-3 class is the eighth norm-2 step at conductor 1. The resulting six predicted components exceed the earlier mixed cases in the recorded predictor panel. The first permitted field prime is p=23743, with u=16 and $u^2-4p=-2631\cdot 6^2$.

The independent two-sheet gate found 48, 48, 144, and 144 rational CM j values at the four conductors, each with exactly two rational X_0(11) lifts and the expected trace. Under the checked-Sage launcher, the copied regularized producer completed under its 600-second cap and saved the full point and relation matrix, launch receipt, stdout and stderr. The checked runtime, exact source hashes, and gate status are retained alongside the protocol.

The independent Sage audit reconstructs all 768 points, checks each distinct j value's endomorphism-order conductor, compares all 1,536 point-degree rational-root counts with the classical modular polynomials including multiplicities, and replays all 576 saved relation rows and group sums. It obtains the full and nonfloor ranks over QQ, 23, and 1000000007. Its degree-subset ranks are 384/378 for degree 2 and 192/144 for degree 3, where each pair is full/nonfloor rank.

The floor audit stores every signed source-row coefficient and floor vector. The independent integer validator reconstructs all 54 vectors from the raw matrix, checks both disjoint floor coverings, finds eight mixed pair overlaps per cycle, and obtains witness/full/nonfloor ranks 54/480/426 over QQ, 23, and 1000000007. Since the 54 integer vectors lie in the floor kernel, are independent, and the exact rank difference is 54, they form a basis of this saved floor-supported row space.

Proof map. The pinned Lean module proves MixedSix.local_mixed_zero and MixedSix.six_by_eight_mixed_zero: given both opposite-sign coordinate equations on each cycle–pair overlap, all six cycle and 48 pair coefficients in a vanishing combination are zero. lake env lean MixedSix.lean built with Lean v4.35.0-rc4 and Mathlib revision 021c46bdd042f67b538b9f9871d556a93713a9a9; its #print axioms record lists only propext, Classical.choice, and Quot.sound. The integer validator supplies the concrete equations and the exact matrix ranks externally. Formalizing the full 768-column row identities and class-action-to-rational-X_0(11) correspondence in Lean remains open; the theorem is the conditional structural step, not a replacement for those bridges. lean4checker --fresh was unavailable on this host.

The first gate-wrapper invocation stopped before launching Sage because a selector file named select.py shadowed Python's standard select module. Renaming it to select_candidate.py resolved that launcher error; the saved gate and producer receipts are from the completed runs. This experiment is a relation-stage diagnostic with candidate_id: null and timing_claim: null. The next proof step is to derive the six mixed incidence components uniformly from the two rational level-11 eigenline labels and the conductor-1 norm-3 class action.

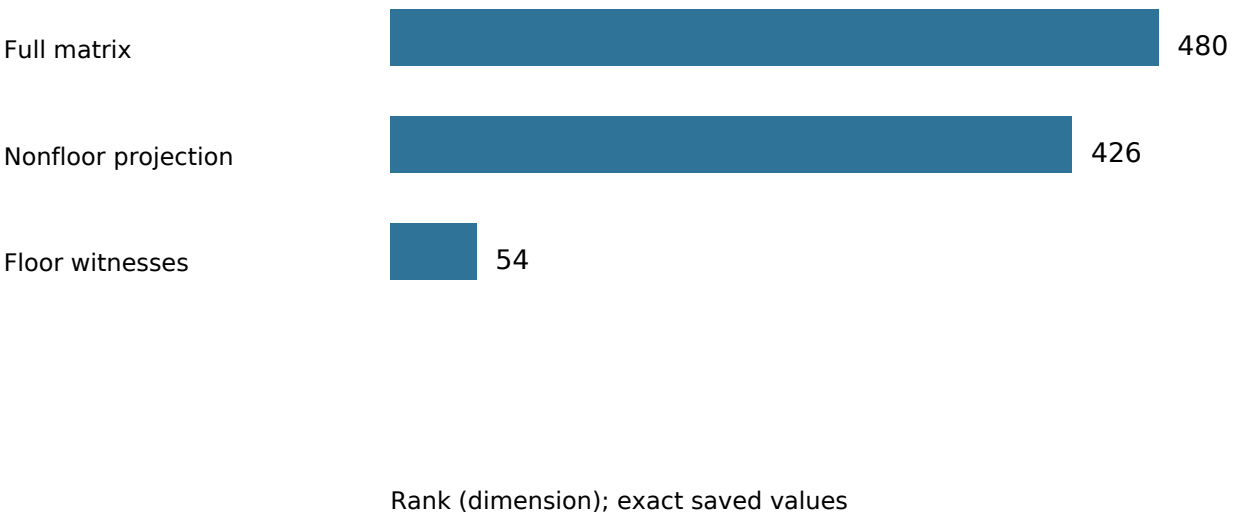
Review provenance and visuals

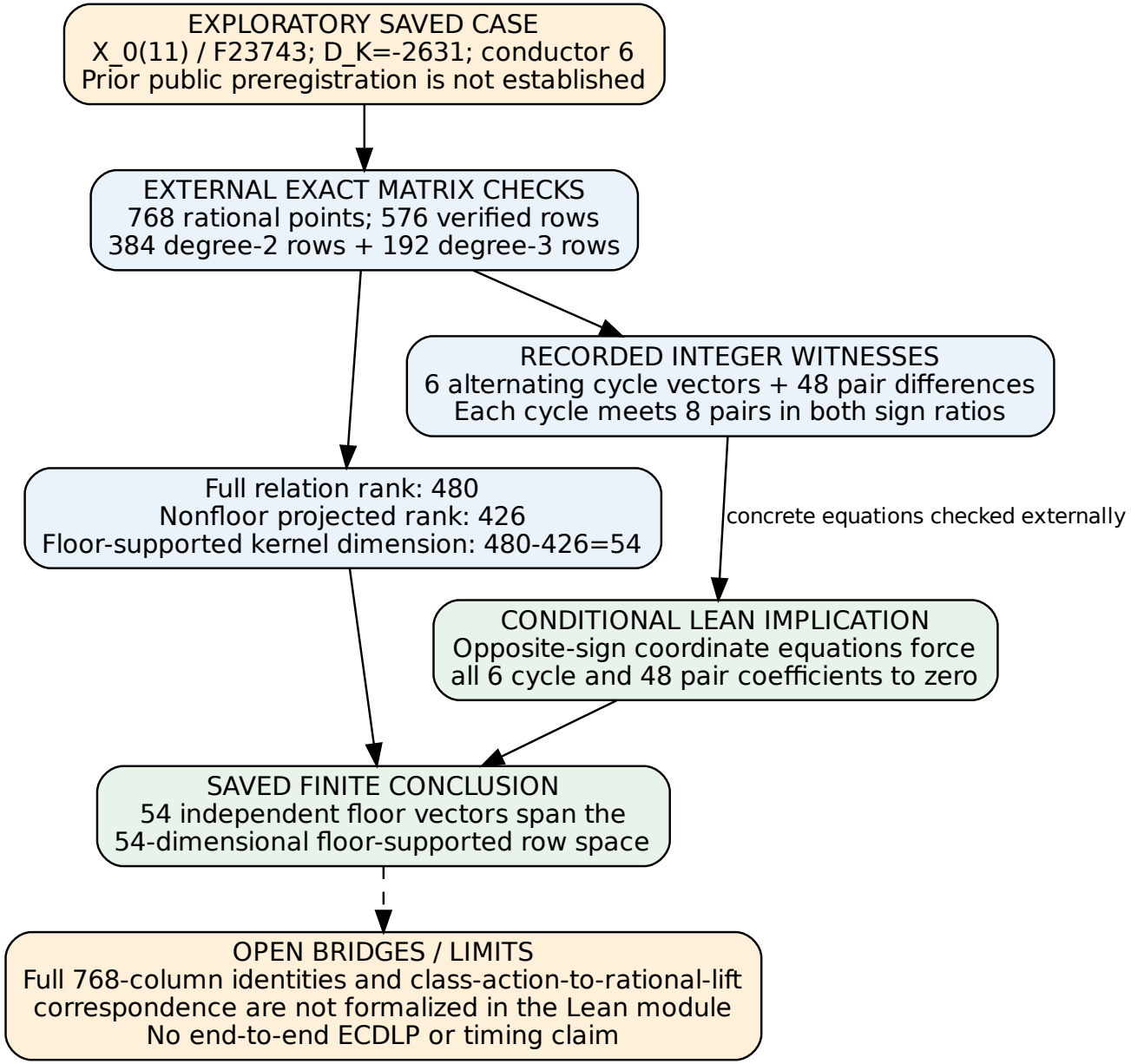
The PDF, evidence diagram with editable Graphviz source, and exact-rank plot present the saved evidence with the exploratory qualification. The review entry point records affected-graph checks and the original source seal's relationship to the current Lean audit version. The version record binds both proof-source byte streams; the statements and proof bodies are unchanged, but visibility and axiom-print placement differ. The original seal, protocols, reports and receipts remain intact. The additive review checker validates the historical bytes and the explicitly pinned current version.

The volcano census and ECDLP figures, D39 rank/control diagrams and associated curve catalog were checked. Their Koblitz descendant and F367 source data do not change. This diagram introduces no new curve, route, subgroup-transport claim or ISO1 admission. Exact saved ranks have no statistical uncertainty; no new Sage/Lean run or timing measurement was executed for this review.

Saved exact ranks

Source: validation.json. The exact rational ranks also agree modulo 23 and 1000000007. These are finite relation-space dimensions, not timing or end-to-end ECDLP measurements. Statistical uncertainty does not apply to these exact saved ranks.




EXPLORATORY SAVED CASE
X_0(11) / F23743; D_K=-2631; conductor 6
Prior public preregistration is not established
EXTERNAL EXACT MATRIX CHECKS
768 rational points; 576 verified rows
384 degree-2 rows + 192 degree-3 rows
RECORDED INTEGER WITNESSES
6 alternating cycle vectors + 48 pair differences
Each cycle meets 8 pairs in both sign ratios
Full relation rank: 480
Nonfloor projected rank: 426
Floor-supported kernel dimension: 480-426=54
concrete equations checked externally
CONDITIONAL LEAN IMPLICATION
Opposite-sign coordinate equations force
all 6 cycle and 48 pair coefficients to zero
SAVED FINITE CONCLUSION
54 independent floor vectors span the
54-dimensional floor-supported row space
OPEN BRIDGES / LIMITS
Full 768-column identities and class-action-to-rational-lift
correspondence are not formalized in the Lean module
No end-to-end ECDLP or timing claim